Lecture 12 2025-10-09

p zero of <u>multiplicity m</u> if $f(x) = (x-p)^m q(x)$ where $\lim_{x\to p} q(x) \neq 0$
& q continuous

Thm 2.12 $f \in C^m([a,b])$. f has a zero of multiplicity m at $x=p$ $\iff$ $f^{(j)}(p) = 0$ $0 \le j \le m-1$
$f^{(m)}(p) \neq 0$

"$\Leftarrow$" ✓ (last class)

"$\Rightarrow$" • If $f^{(j)}(p) = 0$ for $0 \le j \le m$, then $q(p) = \frac{1}{m!} f^{(m)}(p) = 0$
$\Rightarrow$ p is not a zero of multiplicity m

• If $n < m$ and $f^{(j)}(p) = 0$ for $0 \le j \le n-1$ but
$f^{(n)}(p) \neq 0$
then $\frac{f(x)}{(x-p)^n} \to \frac{1}{n!} f^{(n)}(p) \neq 0$ $x \to p$

$$q(x) = (x-p)^{n-m} \frac{f(x)}{(x-p)^n} \xrightarrow{x\to p^{\pm}} \pm\infty$$
$$\left(\frac{f(x)}{(x-p)^m}\right)$$

so q is not continuous at $x=p$.
Contradiction ∎

Homework problem: Show $\frac{d}{dx}\left(\frac{f(x)}{f'(x)}\right) = 1 - \frac{f(x)f''(x)}{(f'(x))^2} \xrightarrow{x\to p} 1 - \frac{m(m-1)}{m^2} = \frac{1}{m}$
if p is a zero of multiplicity m
(Hint: use Taylor expansion)

So if we know m, we can restore quadratic convergence via
$g(x) = x - \underline{mf(x)}$ Newton-m

$$f'(x)$$

$$\left(p_{n+1} = p_n - \frac{m f(p_n)}{f'(p_n)} \quad \text{update formula} \right)$$

Otherwise (don't know m) you can do plain Newton on $\mu(x) = \frac{f(x)}{f'(x)}$. By Homework problem $\mu'(x) \to \frac{1}{m} \neq 0$ as $x \to p$

so Newton should converge quadratically for μ

Iteration: $p_{n+1} = p_n - \frac{\mu(p_n)}{\mu'(p_n)} = p_n - \frac{f(p_n) f'(p_n)}{f'(p_n)^2 - f(p_n) f''(p_n)}$

modified Newton

2.5 Accelerating convergence

Aitken's Δ^2-method: Can be used to accelerate convergence of a sequence that is linearly convergent regardless of origin or application

Recall: $p_n \to p$ linearly if $\left| \frac{p_{n+1} - p}{p_n - p} \right| \to \lambda < 1$ as $n \to \infty$

Motivation: For large n $\frac{p_{n+1} - p}{p_n - p} \approx \lambda \approx \frac{p_{n+2} - p}{p_{n+1} - p}$ (assume same signs for simplicity)

Let $\hat{p}$ solve $\frac{p_{n+1} - \hat{p}}{p_n - \hat{p}} = \frac{p_{n+2} - \hat{p}}{p_{n+1} - \hat{p}}$

Hope: $\hat{p}$ is closer to p than p_n, p_{n+1}, p_{n+2}

$$(p_{n+1}-\hat{p})^2 = (p_n-\hat{p})(p_{n+2}-\hat{p})$$

$$\Longleftrightarrow \quad p_{n+1}^2 - 2\hat{p}p_{n+1} + \hat{p}^2 = p_n p_{n+2} - \hat{p}(p_n + p_{n+2}) + \hat{p}^2$$

$$\Longleftrightarrow \quad p_{n+1}^2 - p_n p_{n+2} = 2\hat{p}p_{n+1} - \hat{p}(p_n + p_{n+2}) = \hat{p}(-p_{n+2} + 2p_{n+1} - p_n)$$

$$\Longleftrightarrow \quad \hat{p} = \frac{p_n p_{n+2} - p_{n+1}^2}{p_{n+2} - 2p_{n+1} + p_n} \qquad (*)$$

$$= \frac{p_n p_{n+2} - p_{n+1}^2 + p_n^2 - p_n^2 + 2p_n p_{n+1} - 2p_n p_{n+1}}{p_{n+2} - 2p_{n+1} + p_n}$$

$$= \frac{p_n(p_{n+2} - 2p_{n+1} + p_n) - (p_{n+1}^2 + p_n^2 - 2p_n p_{n+1})}{p_{n+2} - 2p_{n+1} + p_n}$$

$$= p_n - \frac{(p_{n+1} - p_n)^2}{p_{n+2} - 2p_{n+1} + p_n}$$

Define <u>forward differences</u>:

$$\Delta p_n = p_{n+1} - p_n$$

$$\Delta^2 p_n = (p_{n+2} - p_{n+1}) - (p_{n+1} - p_n)$$

$$\Delta^k p_n = \Delta^{k-1} p_{n+1} - \Delta^{k-1} p_n$$

$$\hat{p}_n = p_n - \frac{(\Delta p_n)^2}{\Delta^2 p_n}$$

<u>Aitken Δ^2-method</u>

<u>Hope</u>: $\{\hat{p}_n\}_{n=1}^{\infty}$ converges faster to p than $\{p_n\}_{n=1}^{\infty}$

<u>Theorem 2.14</u> Suppose $\{p_n\}_{n=1}^{\infty}$ converges linearly to p with $\lim_{n\to\infty} \frac{p_{n+1}-p}{p_n-p} < 1$.

Then Aitken's Δ^2-method converges faster to p than $\{p_n\}_{n=1}^{\infty}$ in the sense that

$$\lim_{n\to\infty} \frac{\hat{p}_n - p}{p_n - p} = 0$$

Proof: $\varepsilon_n = p_n - p \qquad \hat{\varepsilon}_n = \hat{p}_n - p$

$\Rightarrow$ WTS: $\dfrac{\hat{\varepsilon}_n}{\varepsilon_n} \to 0$ as $n \to \infty$

Note: $\hat{\varepsilon}_n = \varepsilon_n - \dfrac{(\Delta\varepsilon_n)^2}{\Delta^2 \varepsilon_n}$, because $\Delta\varepsilon_n = \Delta(p_n - p)$

$= (p_{n+1} - p) - (p_n - p)$

$= p_{n+1} - p_n$

$= \Delta p_n$

$$= \frac{\varepsilon_n \varepsilon_{n+2} - \varepsilon_{n+1}^2}{\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n}$$

↗ follows from formula (*) above

So $$\frac{\hat{\varepsilon}_n}{\varepsilon_n} = \frac{1}{\varepsilon_n}\left(\frac{\varepsilon_n \varepsilon_{n+2} - \varepsilon_{n+1}^2}{\varepsilon_{n+2} - 2\varepsilon_{n+1} + \varepsilon_n}\right) = \frac{\frac{\varepsilon_{n+2}}{\varepsilon_n} - \left(\frac{\varepsilon_{n+1}}{\varepsilon_n}\right)^2}{\frac{\varepsilon_{n+2}}{\varepsilon_n} - \frac{2\varepsilon_{n+1}}{\varepsilon_n} + 1} = (\Delta)$$

We know $\dfrac{\varepsilon_{n+1}}{\varepsilon_n} = \dfrac{p_{n+1} - p}{p_n - p} \longrightarrow \lambda < 1$ as $n \to \infty$

(Assumption of linear convergence)

$$\frac{\varepsilon_{n+2}}{\varepsilon_n} = \frac{\varepsilon_{n+2}}{\varepsilon_{n+1}} \frac{\varepsilon_{n+1}}{\varepsilon_n} \longrightarrow \lambda^2 \text{ as } n \to \infty$$

$$(\Delta) \longrightarrow \frac{\lambda^2 - \lambda^2}{\lambda^2 - 2\lambda + 1} = \frac{\lambda^2 - \lambda^2}{\underbrace{(\lambda - 1)^2}_{\neq 0}} = 0$$

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(Example 1 in book: $p_n = \cos\frac{1}{n}$ has $\lambda = 1$, so this Theorem does not apply)

One issue with Aitken's Δ^2 method: We are not making use of $\hat{p}_n$ in subsequent iterations.